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Linear Algebra - Week 6

Fall 2026

This document presents material that was used in Practical Sessions

It has been prepared by G. Neofotistos, in conversations with OpenAI model 6.1 Sol.

The document contains:

I) Short tutorial on LU, LDU, and normalized U.

II) Short tutorial on Gaussian elimination (Week 5 topic).

III) Detailed example on how to find the inverse of a matrix by using the identity matrix in an "augmented-type" matrix.

I. Short tutorial on LU, LDU, and normalized U.

1. LU factorization

Gaussian elimination transforms a matrix A into an upper triangular matrix U. The elimination multipliers are stored in a lower triangular matrix L:

$$\boxed{A = LU.}$$

In the usual convention, L has ones on its diagonal.

Consider

$$A = \begin{pmatrix} 2 & 4 & 2 \\ 4 & 11 & 7 \\ 2 & 7 & 8 \end{pmatrix}.$$

First eliminate the entries below the first pivot, 2:

$$R_2 \leftarrow R_2 - 2R_1, \qquad R_3 \leftarrow R_3 - R_1.$$

This gives

$$\begin{pmatrix} 2 & 4 & 2 \\ 0 & 3 & 3 \\ 0 & 3 & 6 \end{pmatrix}.$$

Next eliminate below the second pivot, 3:

$$R_3 \leftarrow R_3 - R_2.$$

Thus

$$U = \begin{pmatrix} 2 & 4 & 2 \\ 0 & 3 & 3 \\ 0 & 0 & 3 \end{pmatrix}.$$

The multipliers were $m_{21} = 2$, $m_{31} = 1$, and $m_{32} = 1$, so

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}.$$

Therefore,

$$\boxed{A = \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}}_{L}\underbrace{\begin{pmatrix} 2 & 4 & 2 \\ 0 & 3 & 3 \\ 0 & 0 & 3 \end{pmatrix}}_{U}.}$$

Notice that L **stores the multipliers with their original signs**: the row operation subtracts $2R_1$, but $L_{21} = +2$.

2. Normalized U

To normalize U, divide **each row by its diagonal entry**, making every diagonal entry equal to 1.

Extract the pivots into

$$D = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix}.$$

Then define

$$\widehat{U} = D^{-1}U = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

Hence

$$\boxed{U = D\widehat{U}.}$$

The order matters: **multiplication by D on the left scales rows**. This normalization means a unit diagonal; it does not mean that the rows have length 1.

3. LDU factorization

Substitute $U = D\widehat{U}$ into $A = LU$:

$$\boxed{A = LD\widehat{U}.}$$

For our example,

$$A = \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}}_{L} \underbrace{\begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix}}_{D} \underbrace{\begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}}_{\widehat{U}}.$$

Many textbooks rename $\widehat{U}$ as U and write simply $A = LDU$. Always check the convention:

Factorization	L	D	Upper triangular factor
$A = LU$	Unit diagonal	Not shown separately	Diagonal contains the pivots
$A = LD\widehat{U}$	Unit diagonal	Contains the pivots	Unit diagonal

4. Solving a system using the factors

Suppose

$$A\mathbf{x} = \begin{pmatrix} 8 \\ 22 \\ 17 \end{pmatrix}.$$

Using $A = LU$, solve two triangular systems.

First, **forward substitution**:

$$L\mathbf{y} = \mathbf{b} \quad \Longrightarrow \quad \begin{cases} y_1 = 8, \\ 2y_1 + y_2 = 22, \\ y_1 + y_2 + y_3 = 17. \end{cases}$$

Therefore,

$$\mathbf{y} = \begin{pmatrix} 8 \\ 6 \\ 3 \end{pmatrix}.$$

Next, **back substitution**:

$$U\mathbf{x} = \mathbf{y} \quad \Longrightarrow \quad \begin{cases} 2x_1 + 4x_2 + 2x_3 = 8, \\ 3x_2 + 3x_3 = 6, \\ 3x_3 = 3. \end{cases}$$

Working upward gives

$$\boxed{x_1 = x_2 = x_3 = 1.}$$

With LDU, the same calculation becomes three steps:

$$L\mathbf{y} = \mathbf{b}, \qquad D\mathbf{z} = \mathbf{y}, \qquad \widehat{U}\,\mathbf{x} = \mathbf{z}.$$

When row exchanges are necessary, the standard form is $PA = LU$, or $PA = LD\widehat{U}$, where P records the exchanges. Extracting and dividing by the pivots as above requires them to be nonzero.

II. Tutorial on soving 3x3 systems by Gaussian elimination

The key method is **Gaussian elimination**: simplify the system until the last equation contains only one unknown, then solve upward by **back substitution**. Matrix notation helps organize the calculation and explain why it works.

This tutorial uses original examples centered on three equations in three unknowns. The coverage follows the geometry and elimination topics assigned to §§2.1–2.3 of *Introduction to Linear Algebra*, 4th ed., in MIT's reading guide, and the elimination, matrix multiplication, and triangular-factorization topics of the requested sections of *Linear Algebra and Its Applications*. MIT OpenCourseWare (https://www.ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/pages/readings/?utm_source=chatgpt.com)

1. What does solving a 3×3 system mean?

Consider

$$\begin{aligned} 2x + y - z &= 1, \\ 4x + 3y + z &= 13, \\ -2x + y + 2z &= 6. \end{aligned} \tag{1}$$

A **solution** is a triple (x, y, z) that satisfies all three equations simultaneously.

There are three useful ways to view this system.

The equation—or row—picture

Each equation represents a plane in three-dimensional space. A solution lies on all three planes.

Their common intersection may be:

- One point: a unique solution.
- A line or a plane: infinitely many solutions.
- Empty: no solution.

Elimination changes the equations describing the planes, but preserves their common intersection.

The matrix picture

Write the system as

$$A\mathbf{x} = \mathbf{b},$$

where

$$A = \begin{bmatrix} 2 & 1 & -1 \\ 4 & 3 & 1 \\ -2 & 1 & 2 \end{bmatrix}, \qquad \mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} 1 \\ 13 \\ 6 \end{bmatrix}.$$

Here A is the **coefficient matrix**, $\mathbf{x}$ is the unknown vector, and $\mathbf{b}$ is the right-hand-side vector.

Matrix–vector multiplication reproduces the equations:

$$A\mathbf{x} = \begin{bmatrix} 2x + y - z \\ 4x + 3y + z \\ -2x + y + 2z \end{bmatrix}.$$

The column picture

The same equation can be written

$$x \begin{bmatrix} 2 \\ 4 \\ -2 \end{bmatrix} + y \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} + z \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 13 \\ 6 \end{bmatrix}.$$

Thus, solving the system means finding the **weights** x, y, z that combine the columns of A to produce $\mathbf{b}$.

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2. Why are row operations allowed?

Three operations preserve the solution set:

Operation	Notation	Reason
Exchange two equations	$R_i \leftrightarrow R_j$	Their order does not affect the conditions
Multiply an equation by a nonzero number	$R_i \leftarrow cR_i$, $c \neq 0$	The new equation is equivalent to the old one
Add a multiple of another equation	$R_i \leftarrow R_i + cR_j$	The operation can be reversed

For example, replacing equation 2 by “equation 2 minus twice equation 1” is reversible: add twice equation 1 to recover the original equation 2.

Apply every operation to the entire equation, including its right-hand side.

3. Gaussian elimination: a complete worked example

Return to system (1):

$$
\begin{aligned}
2x + y - z &= 1,\\
4x + 3y + z &= 13,\\
-2x + y + 2z &= 6.
\end{aligned}
$$

Our goal is an **upper triangular system**:

$$
\begin{aligned}
u_{11}x + u_{12}y + u_{13}z &= c_1,\\
u_{22}y + u_{23}z &= c_2,\\
u_{33}z &= c_3.
\end{aligned}
$$

Step 1: Eliminate x from equation 2

The coefficient of x in equation 1 is 2. This is the first **pivot**.

The elimination multiplier is

$$\ell_{21} = \frac{4}{2} = 2.$$

Perform

$$R_2 \leftarrow R_2 - 2R_1.$$

Explicitly,

$$(4x + 3y + z) - 2(2x + y - z) = 13 - 2(1),$$

so

$$y + 3z = 11.$$

Step 2: Eliminate x from equation 3

The multiplier is

$$\ell_{31} = \frac{-2}{2} = -1.$$

Perform

$$R_3 \leftarrow R_3 - (-1)R_1 = R_3 + R_1.$$

This gives

$$(-2x + y + 2z) + (2x + y - z) = 6 + 1,$$

hence

$$2y + z = 7.$$

The system is now

$$
\begin{aligned}
2x + y - z &= 1,\\
y + 3z &= 11,\\
2y + z &= 7.
\end{aligned}
$$

Step 3: Eliminate y from equation 3

The second pivot is 1, the coefficient of y in the updated equation 2.

The multiplier is

$$\ell_{32} = \frac{2}{1} = 2.$$

Perform

$$R_3 \leftarrow R_3 - 2R_2.$$

Then

$$(2y + z) - 2(y + 3z) = 7 - 2(11),$$

giving

$$-5z = -15.$$

We have reached

$$\boxed{\begin{aligned} 2x + y - z &= 1, \\ y + 3z &= 11, \\ -5z &= -15. \end{aligned}}$$

The pivots are

$$\boxed{2, \quad 1, \quad -5.}$$

Notice that the second and third pivots come from the **updated equations**, not necessarily from the original diagonal entries.

4. Back substitution

Solve from the bottom upward.

From equation 3:

$$-5z = -15 \quad \Rightarrow \quad z = 3.$$

From equation 2:

$$y + 3(3) = 11 \quad \Rightarrow \quad y = 2.$$

From equation 1:

$$2x + 2 - 3 = 1 \quad \Rightarrow \quad x = 1.$$

Therefore,

$$\boxed{(x, y, z) = (1, 2, 3).}$$

Check in the **original** equations:

$$\begin{aligned} 2(1) + 2 - 3 &= 1, \\ 4(1) + 3(2) + 3 &= 13, \end{aligned}$$

5. The same calculation with an augmented matrix

The **augmented matrix** places the coefficients and right-hand sides together:

$$[A \mid \mathbf{b}] = \left[\begin{array}{ccc|c} 2 & 1 & -1 & 1 \\ 4 & 3 & 1 & 13 \\ -2 & 1 & 2 & 6 \end{array}\right].$$

Apply

$$R_2 \leftarrow R_2 - 2R_1, \qquad R_3 \leftarrow R_3 + R_1 :$$

$$\left[\begin{array}{ccc|c} 2 & 1 & -1 & 1 \\ 0 & 1 & 3 & 11 \\ 0 & 2 & 1 & 7 \end{array}\right].$$

Next,

$$R_3 \leftarrow R_3 - 2R_2 :$$

$$\left[\begin{array}{ccc|c} 2 & 1 & -1 & 1 \\ 0 & 1 & 3 & 11 \\ 0 & 0 & -5 & -15 \end{array}\right].$$

Thus,

$$A\mathbf{x} = \mathbf{b} \quad \longrightarrow \quad U\mathbf{x} = \mathbf{c},$$

with

$$U = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 1 & 3 \\ 0 & 0 & -5 \end{bmatrix}, \qquad \mathbf{c} = \begin{bmatrix} 1 \\ 11 \\ -15 \end{bmatrix}.$$

The unknown vector stays the same; elimination changes the coefficient matrix and the right-hand side together.

6. Elimination itself is matrix multiplication

A row operation can be represented by an **elimination matrix**.

For $R_2 \leftarrow R_2 - 2R_1$,

use

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Multiplying on the left gives

$$E_{21}A = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 1 & 3 \\ -2 & 1 & 2 \end{bmatrix}, \qquad E_{21}\mathbf{b} = \begin{bmatrix} 1 \\ 11 \\ 6 \end{bmatrix}.$$

The second row of $E_{21}A$ is

$$-2(\text{row 1 of } A) + (\text{row 2 of } A).$$

For the other two operations,

$$E_{31} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}, \qquad E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix}.$$

The complete elimination is

$$\boxed{E_{32}E_{31}E_{21}A = U}$$

and

$$\boxed{E_{32}E_{31}E_{21}\mathbf{b} = \mathbf{c}.}$$

The rightmost operation acts first.

Matrix multiplication rules that matter here

For a product BC,

$$(BC)_{ij} = \sum_k b_{ik}c_{kj}.$$

In words: **row i of the first matrix dotted with column j of the second matrix**.

Multiplication is associative:

$$(BC)D = B(CD).$$

It is generally not commutative:

$$BC \neq CB.$$

This matters because applying row operations in a different order can change the result.

7. What if a pivot is zero?

A zero in a proposed pivot position does **not** automatically mean that the system has no unique solution.

Consider

$$\begin{aligned} y + z &= 5, \\ x + y + z &= 6, \\ 2x + 3y + 4z &= 20. \end{aligned}$$

Its augmented matrix is

$$\left[\begin{array}{ccc|c} 0 & 1 & 1 & 5 \\ 1 & 1 & 1 & 6 \\ 2 & 3 & 4 & 20 \end{array}\right].$$

The first proposed pivot is zero. Exchange rows 1 and 2:

$$R_1 \leftrightarrow R_2.$$

Then

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 1 & 5 \\ 2 & 3 & 4 & 20 \end{array}\right].$$

Apply

$$R_3 \leftarrow R_3 - 2R_1 : \qquad \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 1 & 5 \\ 0 & 1 & 2 & 8 \end{array}\right].$$

Then

$$R_3 \leftarrow R_3 - R_2 : \qquad \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 1 & 5 \\ 0 & 0 & 1 & 3 \end{array}\right].$$

Back substitution gives

$$\boxed{z = 3, \quad y = 2, \quad x = 1.}$$

A row exchange can be represented by a **permutation matrix**. For exchanging rows 1 and 2:

$$P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

The reordered system is

$$PA\mathbf{x} = P\mathbf{b}.$$

For numerical computation, **partial pivoting** selects the largest absolute entry available in the current column as the pivot. This helps control the elimination multipliers.

8. How elimination reveals the number of solutions

Consider the family

$$\begin{aligned} x + y + z &= 6, \\ 2x + 3y + 4z &= 20, \\ 3x + 4y + 5z &= k. \end{aligned}$$

Apply

$$R_2 \leftarrow R_2 - 2R_1, \qquad R_3 \leftarrow R_3 - 3R_1.$$

This produces

$$\begin{aligned} x + y + z &= 6, \\ y + 2z &= 8, \\ y + 2z &= k - 18. \end{aligned}$$

Subtract equation 2 from equation 3:

$$0 = k - 26.$$

If $k \neq 26$: no solution

For example, $k = 27$ produces

$$0 = 1,$$

which is impossible.

The original equations are **inconsistent**.

If $k = 26$: infinitely many solutions

The last equation becomes

$$0 = 0.$$

It adds no restriction. Let the **free variable** be $z = t$. Then

$$y = 8 - 2t,$$

and

$$x = 6 - y - z = t - 2.$$

Therefore,

$$\boxed{(x, y, z) = (t - 2,\ 8 - 2t,\ t), \qquad t \in \mathbb{R}.}$$

Result after elimination	Meaning for a 3×3 system
Three pivots in the coefficient matrix	Unique solution
A row $[0\ 0\ 0 \mid d]$, $d \neq 0$	No solution
Fewer than three coefficient pivots, with no contradictory row	Infinitely many solutions

A zero row alone does not guarantee consistency: all remaining rows must also be checked.

9. Triangular factorization: $A = LU$

Elimination also provides a reusable description of the coefficient matrix.

For our first example, the multipliers were

$$\ell_{21} = 2, \qquad \ell_{31} = -1, \qquad \ell_{32} = 2.$$

Place them below the diagonal in

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 2 & 1 \end{bmatrix}.$$

Then

$$\boxed{A = LU,}$$

where

$$\begin{bmatrix} 2 & 1 & -1 \\ 4 & 3 & 1 \\ -2 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 & -1 \\ 0 & 1 & 3 \\ 0 & 0 & -5 \end{bmatrix}.$$

Here L is **lower triangular**, and U is **upper triangular**.

The signs in L are the multipliers themselves. The elimination operation uses their negatives:

$$R_i \leftarrow R_i - \ell_{ij} R_j.$$

Solving with LU

Instead of solving $LU\mathbf{x} = \mathbf{b}$ in one step, introduce $\mathbf{c}$ and solve $L\mathbf{c} = \mathbf{b}, \qquad U\mathbf{x} = \mathbf{c}$.

First, use **forward substitution**:

$$\begin{aligned} c_1 &= 1, \\ 2c_1 + c_2 &= 13 \qquad \Rightarrow c_2 = 11, \\ -c_1 + 2c_2 + c_3 &= 6 \qquad \Rightarrow c_3 = -15. \end{aligned}$$

Next, solve $U\mathbf{x} = \mathbf{c}$ by back substitution, obtaining $(1, 2, 3)$.

The advantage appears when several systems share the same A but have different right-hand sides: factor A once and reuse L and U.

When row exchanges are needed, the standard factorization is

$$\boxed{PA = LU.}$$

10. A practical procedure for any 3×3 system

1. Write the augmented matrix, keeping the variables in the same order in every row.
2. Choose a nonzero first pivot; exchange rows if necessary.
3. Eliminate the entries below that pivot.
4. Choose the second pivot in the updated matrix.
5. Eliminate the entry below it.
6. Check for a contradiction or free variables.
7. If there are three pivots, use back substitution.
8. Verify the solution in the original system.

For an upper triangular system,

$$
\begin{aligned}
u_{11}x + u_{12}y + u_{13}z &= c_1,\\
u_{22}y + u_{23}z &= c_2,\\
u_{33}z &= c_3,
\end{aligned}
$$

the back-substitution formulas are

$$
\boxed{z = \frac{c_3}{u_{33}}, \qquad y = \frac{c_2 - u_{23}z}{u_{22}}, \qquad x = \frac{c_1 - u_{12}y - u_{13}z}{u_{11}}.}
$$

These formulas require nonzero pivots.

Common mistakes are forgetting to update the right-hand side, using an old row after it has changed, reversing the multiplier ratio, and treating a zero proposed pivot as proof of singularity.

11. Practice exercises

A. Elimination and back substitution

Solve

$$
\begin{aligned}
x + y + z &= 6,\\
2x + 3y + z &= 11,\\
x + 2y + 3z &= 14.
\end{aligned}
$$

Answer: $(x, y, z) = (1, 2, 3)$.

After eliminating x:

$$
y - z = -1, \qquad y + 2z = 8.
$$

Subtracting gives $3z = 9$.

B. A row exchange

Solve

$$
\begin{aligned}
y + 2z &= 8,\\
x + y + z &= 6,\\
2x + 3y + z &= 11.
\end{aligned}
$$

Answer: $(1, 2, 3)$. Exchange the first two rows before eliminating x.

C. Classify the solution set

Determine when the system has no solution or infinitely many solutions:

$$
\begin{aligned}
x + y + z &= 6,\\
2x + 3y + 4z &= 20,\\
3x + 4y + 5z &= k.
\end{aligned}
$$

Answer: No solution for $k \neq 26$; infinitely many for $k = 26$. There is no value of k giving a unique solution.

D. Construct an elimination matrix

Find the matrix representing

$$R_3 \leftarrow R_3 - 4R_1.$$

Answer:

$$E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -4 & 0 & 1 \end{bmatrix}.$$

III. Example on how to find the inverse of a matrix by using the identity matrix in an "augmented-type" matrix

To find the inverse of a square matrix A, place the identity matrix beside it and apply row operations until the left side becomes the identity:

$$\boxed{[A \mid I] \longrightarrow [I \mid A^{-1}].}$$

This method is called **Gauss–Jordan elimination**.

1. The theory

The inverse of an $n \times n$ matrix A is a matrix A^{-1} satisfying

$$AA^{-1} = A^{-1}A = I_n.$$

Each elementary row operation is equivalent to multiplication on the left by an elementary matrix. If a sequence of row operations transforms A into I, then

$$E_k \cdots E_2 E_1 A = I.$$

Therefore,

$$E_k \cdots E_2 E_1 = A^{-1}.$$

Applying those same operations to the identity matrix produces

$$E_k \cdots E_2 E_1 I = A^{-1}.$$

That is why

$$[A \mid I] \longrightarrow [I \mid A^{-1}].$$

Every row operation must act on the entire augmented row, on both sides of the vertical line.

The permitted operations are:

- Exchange two rows.
- Multiply a row by a nonzero number.
- Add a multiple of one row to another.

2. A complete 3×3 example

Find the inverse of

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}.$$

Step 1: Form $[A \mid I]$.

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{array}\right].$$

Step 2: Eliminate the entry below the first pivot.

Apply

$$R_2 \leftarrow R_2 - R_1.$$

We obtain

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{array}\right].$$

Step 3: Eliminate the entry below the second pivot.

Apply

$$R_3 \leftarrow R_3 - R_2.$$

This gives

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1 & -1 & 1 \end{array}\right].$$

The left side is now upper triangular. To find the inverse, we continue until it becomes I.

Step 4: Eliminate the entry above the third pivot.

Apply

$$R_2 \leftarrow R_2 - R_3.$$

Then

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -2 & 2 & -1 \\ 0 & 0 & 1 & 1 & -1 & 1 \end{array}\right].$$

Step 5: Eliminate the entry above the second pivot.

Apply

$$R_1 \leftarrow R_1 - R_2.$$

We obtain

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 3 & -2 & 1 \\ 0 & 1 & 0 & -2 & 2 & -1 \\ 0 & 0 & 1 & 1 & -1 & 1 \end{array}\right].$$

The left side is the identity matrix, so the right side is the inverse:

$$\boxed{A^{-1} = \begin{bmatrix} 3 & -2 & 1 \\ -2 & 2 & -1 \\ 1 & -1 & 1 \end{bmatrix}.}$$

3. Verify the result

Multiply:

$$AA^{-1} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 3 & -2 & 1 \\ -2 & 2 & -1 \\ 1 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

For example, the entry in row 2, column 1 is

$$1(3) + 2(-2) + 1(1) = 0.$$

4. When does the method fail?

A square matrix has an inverse **only if its left block can be reduced to the identity**.

If elimination produces a row of zeros in the left block, there are too few pivots to form I, and A is **singular**, meaning it has no inverse.

For example,

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 2 & 4 & 0 & 1 \end{array}\right] \xrightarrow{R_2 \leftarrow R_2 - 2R_1} \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 0 & -2 & 1 \end{array}\right].$$

The left block cannot become I_2, so

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

has no inverse.

Gaussian elimination stops at an upper triangular matrix; Gauss–Jordan elimination continues until every pivot is 1 and all entries above and below each pivot are 0.

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